E214 ATLAS

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(Dated: November 10, 2026)

We present an analysis of simulated and recorded proton-proton collision data from the ATLAS experiment, focusing on electromagnetic calorimeter (ECAL) calibration, particle identification, and a measurement of the W-boson mass. First, we validate the electron identification using the ratio of ECAL cluster energy to inner-detector track momentum (E/p), and analyze distinct event topologies to classify an unknown dataset as heavy-quark pair production ($t\bar{t}$ or $b\bar{b}$). The ECAL energy response is subsequently calibrated using the invariant mass spectrum of $Z^0 \to e^+e^-$ decays, yielding a reconstructed Z-boson mass of 91.18(1) GeV, in agreement with the global baseline [1] . Building upon this calibration, the W-boson mass is extracted via the Jacobian-peak method applied to the electron transverse-momentum (p_T^e) distribution in $W \to e\nu$ events. Following a stringent kinematic cut selection and QCD background scaling, the mass is determined using a linear gauge curve constructed from Monte Carlo templates. We report a final W-boson mass of $m_W = 80.83 \pm 0.08$ (stat.) $^{+0.41}_{-0.81}$ (sys.) GeV. This result deviates from the Particle Data Group world average of $m_W^{\mathrm{PDG}} = 80.3692(133)$ GeV [1] by 0.57σ. The measurement is heavily systematic-limited, with the dominant uncertainties arising from low-energy QCD background modeling and kinematic fit-range boundaries.

I. INTRODUCTION

The W and Z^0 bosons are the massive gauge bosons of the electroweak interaction in the Standard Model of particle physics. Their masses are not free parameters but are related through the electroweak mixing angle θ_W [1]. Precision measurements of the W-boson mass therefore provide a stringent consistency test of the Standard Model and constrain potential contributions from physics beyond it.

This report presents an analysis of proton–proton collision data recorded by the ATLAS experiment at the Large Hadron Collider (LHC) at CERN, where beams collide at centre-of-mass energies of up to $\sqrt{s} = 14\,\mathrm{TeV}$ [2]. At these energies, W and Z bosons are copiously produced as real, on-shell particles with sizeable cross-sections, enabling direct studies of their properties.

A. The ATLAS Detector

The ATLAS detector is a general-purpose particle detector with a cylindrical geometry, measuring 46 m in length, 25 m in height and weighing approximately 7000 tonnes [2]. Its layered design ensures that nearly all particle species produced in collisions can be identified and their momenta or energies measured. The main subsystems, ordered from the interaction point outward, are as follows.

The **Inner Detector (ID)** reconstructs the trajectories of charged particles using silicon pixel and strip sensors together with a gas-filled transition radiation tracker. Immersed in a 2 T solenoidal magnetic field aligned with the beam axis, the ID measures the curvature of charged tracks and thereby determines their momenta [2].

The **Electromagnetic Calorimeter (ECAL)** absorbs and measures the energy of electrons, positrons and photons through the development of electromagnetic showers. The ATLAS ECAL employs a liquid-argon sampling technology with a distinctive accordion geometry, in which lead absorber plates and active detection layers are folded to ensure full azimuthal coverage without dead regions [2]. Since the W-mass measurement presented in this report relies on the electron transverse-momentum spectrum, an accurate calibration of the ECAL is a central prerequisite of the analysis.

The **Hadronic Calorimeter (HCAL)** surrounds the ECAL and is designed to fully contain hadronic showers produced by strongly interacting particles. It uses iron absorbers with scintillation-tile readout in the barrel region and liquid-argon technology in the forward sections [2].

The **Muon Spectrometer** forms the outermost layer of the detector. Because muons traverse the calorimeters without forming showers (owing to their large mass relative to the electron), they are the only detectable charged particles that reach this subsystem. Dedicated toroidal magnets provide an independent momentum measurement [2].

The combination of these subsystems yields a characteristic detector-response matrix: electrons produce a track in the ID matched to an ECAL cluster; photons deposit energy in the ECAL without an associated track;

muons leave tracks in both the ID and the muon spectrometer with only minimal calorimeter deposits; and hadrons are absorbed primarily in the HCAL. Neutrinos escape the detector entirely and are inferred from the imbalance of the total transverse momentum, quantified by the missing transverse energy E_T^{miss} [2].

B. Overview of this Report

The analysis proceeds in several stages. Section II collects preparatory calculations on two-body decay kinematics, the reconstruction of the W-boson transverse momentum, and the treatment of correlated fit-parameter uncertainties. Section III presents a study of the ratio of ECAL cluster energy to Inner-Detector track momentum (E/p) for individual electron candidates, serving as an initial check of detector response. In Section IV, a set of mystery event displays is examined and the underlying Standard Model process is identified. Section V describes the iterative calibration of the ECAL energy scale using the well-known $Z^0 \to e^+e^-$ resonance. Finally, Section VI details the extraction of the W-boson mass via the Jacobian peak method applied to the electron transverse-momentum spectrum, including a comprehensive evaluation of systematic and statistical uncertainties.

II. QUESTIONS TO BE ANSWERED BEFORE THE LAB

a. ***Question 6.1.1 [2] : Kinematics of Z^0 decay at rest.*** Consider the two-body decay of a Z^0 boson at rest into an electron-positron pair ($Z^0 \to e^-e^+$). By conservation of total momentum, the initial momentum of zero dictates that the momenta of the electron ($\vec{p}_{e^-}$) and positron ($\vec{p}_{e^+}$) must be equal in magnitude but opposite in direction:

$$\vec{p}_{e^-} + \vec{p}_{e^+} = 0 \quad \Longrightarrow \quad |\vec{p}_{e^-}| = |\vec{p}_{e^+}| \tag{1}$$

Furthermore, since the electron and positron have identical masses and momentum magnitudes, they must carry equal energy ($E_{e^-} = E_{e^+}$). Conservation of energy then requires:

$$M_{Z^0} = E_{e^-} + E_{e^+} = 2E_{e^-} \quad \Longrightarrow \quad E_{e^-} = \frac{M_{Z^0}}{2}\,. \tag{2}$$

Using the relativistic energy-momentum relation, the momentum of the electron can be explicitly calculated:

$$|\vec{p}_{e^-}| = |\vec{p}_{e^+}| = \sqrt{E_{e^-}^2 - m_e^2} = \sqrt{\frac{M_{Z^0}^2}{4} - m_e^2} \tag{3}$$

Because the mass of the Z^0 boson is vastly larger than the electron mass ($M_{Z^0} \gg m_e$), the m_e^2 term is negligible, yielding $|\vec{p}_{e^-}| \approx M_{Z^0}/2$.

b. ***Question 6.1.2 [2] : Kinematics of $\tau^+\tau^-$ production.*** In the center-of-mass (COM) frame, the total initial momentum of the system is zero. Therefore, a produced $\tau^+\tau^-$ pair must be emitted back-to-back with equal momentum magnitudes:

$$\vec{p}_{\tau^-} + \vec{p}_{\tau^+} = 0 \quad \Longrightarrow \quad |\vec{p}_{\tau^-}| = |\vec{p}_{\tau^+}| = p_\tau \tag{4}$$

Given that both τ leptons have an identical mass of $m_\tau \approx 1.777\,\text{GeV}$ [1] , the momentum equality implies they also have equal energies ($E_{\tau^-} = E_{\tau^+}$). Assuming an available COM energy of 5 GeV such that each tau lepton carries $E_\tau = 2.5\,\text{GeV}$, the momentum can be directly computed as:

$$\begin{aligned} p_\tau &= \sqrt{E_\tau^2 - m_\tau^2} \\ &= \sqrt{(2.5\,\text{GeV})^2 - (1.777\,\text{GeV})^2} \approx 1.758\,\text{GeV} \end{aligned} \tag{5}$$

c. ***Question 7.1.1 [2] : Proxy for missing momentum.*** The transverse momentum of the W boson is the vector sum of the transverse momenta of the electron (e) and the neutrino (ν). Since the neutrino is not directly detected, its transverse momentum is represented by the Missing Transverse Energy vector $\vec{E}_T^{\text{miss}}$.

Using the variables provided in the ROOT tree, the calculation is performed as follows: The transverse momentum vector of the W boson, $\vec{p}_{T,W}$, is defined by its x and y components:

- Electron components: `el_px`, `el_py`
- Neutrino (Missing Energy) components: `ptmisx`, `ptmisy`

By conservation of momentum in the transverse plane, the components of the W boson's momentum are the sum of the components of its decay products:

$$p_{x,W} = \texttt{el_px} + \texttt{ptmisx} \tag{6}$$

$$p_{y,W} = \texttt{el_py} + \texttt{ptmisy} \tag{7}$$

The magnitude of the transverse momentum, `ptw`, is the length of this resultant vector:

$$p_{T,W} = \sqrt{(p_{x,W})^2 + (p_{y,W})^2} \tag{8}$$

Substituting the tree variables directly:

$$\texttt{ptw} = \sqrt{(\texttt{el_px} + \texttt{ptmisx})^2 + (\texttt{el_py} + \texttt{ptmisy})^2} \tag{9}$$

d. ***Question 7.1.2 [2] : Decoupling correlated fit parameters.*** When fitting measured data to a standard linear function $y = ax+b$, the best-fit parameters for the slope (a) and intercept (b) are generally correlated. Their uncertainties are described by a covariance matrix containing an off-diagonal term, $\text{cov}(a, b)$. The simple form of Gaussian error propagation is strictly valid only

for uncorrelated parameters. If correlations exist, one must use the fully expanded formula:

$$\sigma_f^2 = \left(\frac{\partial f}{\partial a}\right)^2 \sigma_a^2 + \left(\frac{\partial f}{\partial b}\right)^2 \sigma_b^2 + 2\left(\frac{\partial f}{\partial a}\right)\left(\frac{\partial f}{\partial b}\right)\mathrm{cov}(a,b) \tag{10}$$

To avoid the complexity of the cross-term, one can eliminate the correlation by transforming the coordinate system—which, mathematically, corresponds to diagonalising the covariance matrix.

In experimental practice, this is easily achieved by re-centering the independent variable around the mean of the data points, $\bar{x}$, and fitting the modified function:

$$y = a\,(x - \bar{x}) + b' \tag{11}$$

In this parameterisation, b' represents the value of the fitted function at the center of the data ($x = \bar{x}$) rather than at $x = 0$. A change in the slope a now simply pivots the line around the center of mass of the data, leaving b' largely unaffected. Consequently, the off-diagonal covariance element $\mathrm{cov}(a, b')$ is strongly suppressed (vanishing exactly for equally spaced, equally weighted data). By rendering the parameters statistically independent, the simple, standard uncorrelated error propagation formula becomes a highly accurate approximation.

III. ECAL CLUSTER ENERGY VS. INNER-DETECTOR TRACK MOMENTUM (ASSIGNMENT 2 [2])

For each of the 20 electron events, the transverse momentum $|p_T|$ from the Inner Detector and the ECAL cluster transverse energy E_T was read from the event display. The ratio $E/p \equiv E_T/|p_T|$ and its uncertainty $\sigma(E/p) = (E/p)\,\delta p_T/|p_T|$ are listed in Table III and shown as a histogram in Figure 1. We note that the event display only provided uncertainties in p_T but not E_T. As we didn't find any sensible way to estimate the error in E_T we decided to omit it, noting that $\sigma(E/p) = (E/p)\,\delta p_T/|p_T|$ is only approximately correct.

For relativistic electrons ($m_e \approx 0.5\,\mathrm{MeV} \ll p$) the approximation $E \approx p$ holds, so a perfectly calibrated detector would yield $E/p \approx 1$ for every event. Excluding three outliers (events 2, 4 and 14 given in Table III in Appendix A), the remaining 17 events give $\langle E/p\rangle = 1.13 \pm 0.08$ with a standard deviation of $\sigma_{E/p} = 0.34$. When all 20 events are included, the mean rises to $\langle E/p\rangle = 2.29 \pm 0.72$ with $\sigma_{E/p} = 3.22$, illustrating how strongly the three outliers distort the sample statistics.

The mean without outliers lies slightly above unity, which is consistent with the use of uncalibrated raw ECAL energies. The spread reflects the finite track-momentum resolution of the Inner Detector and the lack of calorimeter calibration.

Events 2, 4 and 14 have anomalously low momenta of $|p_T| \approx 3$–$7\,\mathrm{GeV}$ yet typical ECAL energies of 30–45 GeV, yielding E/p values between 4.5 and 14.6. The most likely cause is that these are muons that passed the cut selection. Otherwise, a poorly reconstructed track, possibly due to few hits or a short visible arc in the Inner Detector could also be the cause.

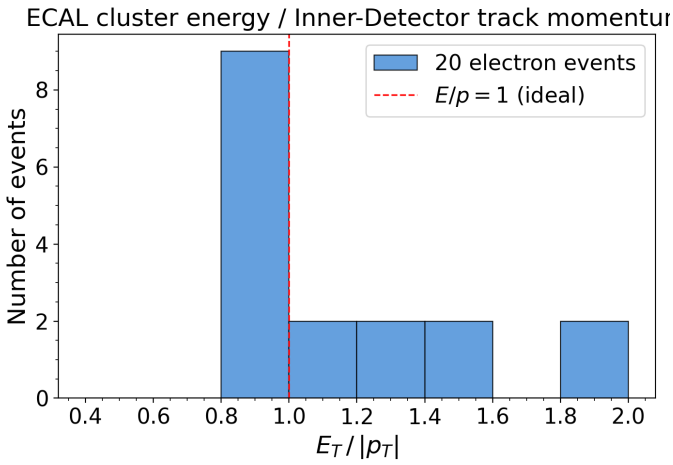


FIG. 1: Distribution of E/p for the 20 electron events. Three outlier events with $E/p > 2$ (events 2, 4, 14) fall outside the displayed range. The dashed red line marks the ideal value $E/p = 1$.

This shows the usefulness of E/p as a quality filter for electron identification.

IV. MYSTERY PARTICLE IDENTIFICATION (ASSIGNMENT 7 [2])

A. Pre-selection and Dataset

The dataset used for this assignment was `mystery-007f.zip`. A pre-selection requiring at least two leptons (e or μ) had already been applied. We examined events 1, 2, 3, 5, 10a, 10b, 12, 45a and 45b, where for numbers 10 and 45 each were given two separate events, which we have labeled a and b respectively. A detailed summary of each event is given in Appendix B.

B. Recurring Patterns

We noticed several features that recurred across the examined events. The lepton content was always either $\mu\mu$ or μe; we never observed ee or more than two leptons. In all cases, the two leptons carried opposite electric charge, as evidenced by opposite p_T signs. In events with two muons, the muons were frequently observed at approximately right angles to each other. The leptons were often accompanied by multiple closely-spaced tracks, and energy deposition was predominantly in the ECAL rather than in the hadronic calorimeter. Finally, E_T^{miss} varied

considerably across the samples (from 2.6 to 164 GeV), which is consistent with a variable number of neutrinos in the final state.

C. Hypothesis Testing

Using the cross-sections of relevant Standard Model processes (see Table IV in Appendix C) as a guide, we considered several candidates that could produce the observed signatures and attempted to exclude as many as possible.

- Z^0 pair production can be excluded because it would require two same-flavour, opposite-sign lepton pairs (e.g. $e^+e^-\mu^+\mu^-$). The frequent observation of mixed-flavour combinations such as μe is incompatible with this process.
- Dijet production with $p_T > 100\,\text{GeV}$ can also be excluded. This topology would require dominant back-to-back hadronic jets, with leptons appearing only as secondary decay products. Only one event in our sample exhibited this structure.
- Single boson production (W or Z) is excluded since the observed track multiplicities and overall event complexity are greater than what a single-boson decay can produce.
- The Higgs decay $H \to Z^0Z^0$ is disfavoured for the same reason as the Z^0 pair production: the $ZZ \to 4\ell$ final state requires same-flavour lepton pairs, which we do not always observe.
- $t\bar{t}$ production remains a viable hypothesis. Each top quark decays via $t \to Wb$, and the subsequent leptonic W decay $W \to \ell\nu$ naturally produces two opposite-charge leptons that can be of different flavour (μe or $\mu\mu$). The b-quarks hadronise into jets with closely-spaced tracks, and the neutrinos from the W decays account for the variable E_T^{miss}.
- Direct $b\bar{b}$ production is also possible, since semileptonic b-decays can yield opposite-charge leptons of different flavour with accompanying hadronic activity.

D. Conclusion

The data are most consistent with heavy-quark pair production, with $t\bar{t}$ and direct $b\bar{b}$ both remaining as viable candidates. While some events exhibit high $\sum E_T$ values that might favour $t\bar{t}$, the sample size and the information accessible through event displays alone are not sufficient to draw a definitive conclusion here. The key signatures supporting the heavy-quark interpretation are: opposite-charge, mixed-flavour dileptons; hadronic jets with closely-spaced tracks and variable E_T^{miss} consistent with neutrinos in the final state. A conclusive distinction between $t\bar{t}$ and direct $b\bar{b}$ would require a more detailed analysis that is beyond the scope of this event-display analysis.

V. CALIBRATION OF ELECTRONS

The electron energy calibration was performed iteratively using $Z^0 \to e^+e^-$ events, fitting the invariant mass spectrum with a Voigt function (Breit–Wigner convolved with a Gaussian detector resolution) to extract the Z peak position in different kinematic regions.

As a first step, we examined the Z peak position separately as a function of the transverse momentum p_T, pseudorapidity η, and azimuthal angle ϕ. No significant dependence on ϕ was observed. As a result, ϕ-dependent corrections were omitted in all subsequent iterations.

Energy-dependent (p_T-binned) and geometry-dependent (η-binned) correction factors were then derived independently, each bringing the fitted Z mass closer to the PDG value of $M_Z = 91.1880(20)\,\text{GeV}$ [1]. After combining both sets of corrections, the energy calibration remained stable across p_T bins. However, the η-dependent calibration degraded, particularly near the calorimeter edges ($|\eta| \gtrsim 2.2$), where correction factors of up to $\sim 8\%$ were required. Additional η-dependent correction factors were therefore applied on top of the combined calibration, with the largest corrections concentrated in the forward regions $|\eta| > 2.2$, while the central region ($|\eta| < 1.5$) required only sub-percent adjustments.

The final correction factors are summarized in Tables V and VI in Appendix D

A. Z^0 Invariant Mass from ATLAS Data

As a validation of the performed ECAL energy calibration, we carried out a global fit[1] of the $Z^0 \to e^+e^-$ invariant-mass spectrum, presented in Figure 2.

The fit yields a Z^0 mass of $91.18(1)\,\text{GeV}$. Using the z-score as a quantitative measure of statistical compatibility to compare the fit result to the current world average of $m_Z = 91.1880(20)\,\text{GeV}$ [1], we obtain a deviation of less than 1σ:

$$z = \frac{|91.1880 - 91.18|}{\sqrt{0.01^2 + 0.0020^2}} = \frac{0.008}{0.0102} \approx 0.78 \qquad (12)$$

indicating no statistically significant deviation. The nominal values and their uncertainties are fully compatible. We note that the analysis software limits the preci-

[1] A global fit refers to fitting the whole data spectrum without applying any kinematic cuts.

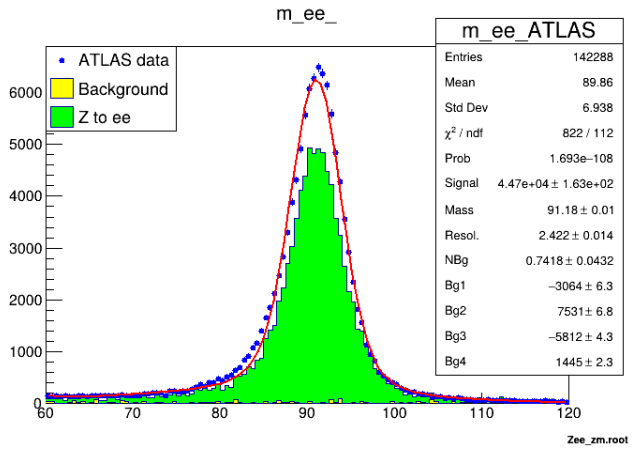


FIG. 2: Invariant-mass distribution of electron–positron pairs in ATLAS data used to validate the ECAL energy calibration (the x-axis is in GeV and the y-axis represents counts). The red curve represents a global Voigt-profile fit to the Z^0 resonance peak. The blue data points represent actual ATLAS data, while the coloured histograms show the stacked simulated signal and background contributions (prior to QCD background scaling). The extracted mass of $m_Z = 91.18(1)\,\text{GeV}$ matches the established world average $m_Z = 91.1880(20)\,\text{GeV}$ [1], confirming that the electron energy calibration is reliable and accurate within the resolution limits of the analysis software.

sion of the fitted mass to the second decimal place, which prevents a more rigorous comparison.

Since the measured Z^0 mass agrees with the world average, we conclude that any residual systematic uncertainty from the ECAL energy calibration is below the resolution accessible in this analysis, though small discrepancies may still exist at higher orders of precision.

VI. W-MASS MEASUREMENT

We measure the mass of the W boson using the Jacobi-peak method [2] applied to the electron transverse-momentum spectrum in the $W \to e\nu$ events recorded by the ATLAS detector. Because the neutrino escapes detection, the full invariant mass of the W boson cannot be reconstructed directly. Instead, we rely on the electron's transverse momentum p_T^e, which exhibits a characteristic Jacobian peak at approximately half the W boson mass m_W. The analysis is structured as follows:

1. Determine the QCD background scaling factor, denoted as α_{QCD}, by visual inspection using a stackplot, as done in Section VI B .

2. Define a cut selection on kinematic variables that maximises the signal-to-background ratio (SBR) , described in Section VI C .

3. Employ the Jacobi-peak method [2] on simulated data sets with well-known W masses to construct a gauge curve as done in Section VI D .

4. Extract the W-boson mass from ATLAS data using the gauge curve, as done in Section VI E .

5. Discuss the various systematic and statistical error influences on the final W mass measurement, detailed in Section VI F .

A. Jacobi Peak Method

In the leptonic decay $W \to e\nu$, the escaping neutrino prevents the direct reconstruction of the W-boson invariant mass. However, the mass can be inferred from the electron's transverse momentum p_T^e distribution in the Laboratory frame. In the W-boson rest frame, the electron and neutrino are emitted back-to-back with a momentum of approximately half the W boson mass m_W. The transverse component of the electron's momentum is given by $p_T^e = \frac{m_W}{2}\sin\theta^*$, where θ^* is the emission angle relative to the beam axis [2].

Assuming the W bosons are unpolarized, the decay is isotropic in the solid angle Ω [3]. This assumption is relevant because any significant polarization would introduce angular dependencies (such as $(1 \pm \cos\theta^*)^2$) [3] that alter the exact shape of the peak. Assuming purely isotropic emission allows the differential cross-section to be transformed from $\cos\theta^*$ to p_T^e strictly via a geometric Jacobian factor [2]:

$$\frac{\mathrm{d}\sigma}{\mathrm{d}p_T^e} = \frac{\mathrm{d}\sigma}{\mathrm{d}\cos\theta^*}\left|\frac{\mathrm{d}\cos\theta^*}{\mathrm{d}p_T^e}\right| \propto \frac{p_T^e}{\sqrt{(m_W/2)^2 - (p_T^e)^2}}\,. \quad (13)$$

This kinematic singularity causes the distribution to peak sharply as p_T^e approaches its kinematic maximum at $m_W/2$. This feature is known as the *Jacobian peak*.

In real collisions, though, this ideal singularity is smeared into a broadened peak by three main effects [2]:

- The intrinsic Breit–Wigner width of the W boson
- The non-zero transverse momentum of the W boson (p_T^W) due to, for instance, initial-state radiation [2]
- The finite detector resolution

Instead of using the peak's absolute maximum, we extract the mass by fitting the falling edge of the distribution to locate its half-maximum point, $H_{1/2}$. The motivation for this approach is that the half-maximum point mathematically remains much more robust against kinematic smearing than the peak maximum itself [2].

In practice, we fit an empirical function to the p_T^e spectrum within a restricted kinematic window, nominally chosen as 30–60 GeV. This fit range is deliberately selected to securely bracket the peak and the full descent

of the trailing edge. The lower bound of 30 GeV avoids heavy low-p_T QCD background contamination, while the upper bound of 60 GeV was found to include enough of the failing tail to ensure stable convergence of the algorithm. An example of these Jacobi-peak fits is depicted in Figure 15 in Appendix F.

B. QCD Background Scaling

Having established the theoretical framework for extracting the mass, we must address the background modeling. The QCD background data set provided with the ATLAS data does not carry a correct absolute normalisation. We therefore determine a QCD scale factor, denoted as α_{QCD}, by comparing the stacked simulation histograms to the ATLAS data in a kinematic region heavily dominated by QCD background.

Our procedure is as follows. We plot the electron transverse momentum p_T^e and manually adjust α_{QCD} until the sum of simulated signal and background histograms matches the ATLAS data points by visual inspection [2]. We choose the electron p_T^e distribution as the primary observable for this scaling because it directly enters the W-mass extraction through the Jacobi-peak position, as detailed in Section VI A.

The resulting nominal scale factor we found is $\alpha_{\text{QCD}} = 0.38$. We note that the uncertainty on this parameter constitutes a major source of systematic error, quantified later in this report in Section VI F 1.

C. Cut Selection

With the background scaled appropriately, we optimize the phase space of our analysis. We apply a cut selection with the primary goal of maximising the signal-to-background ratio (SBR) in the electron p_T spectrum, as initially visualized in Figure 3. The final chosen selection is:

```
njet == 0 && el_etiso < 4 && ptw < 36 && 25 < etmis
```

We motivate each individual criterion below, presenting the respective distributions before the cuts are applied to justify the chosen thresholds.

a. ***Jet multiplicity (`njet == 0`):*** Requiring zero reconstructed jets strongly improves the SBR (see Figure 4). When events with one or more jets are included, the W boson is likely recoiling against emitted gluons due to Initial State Radiation (ISR) [2]. This additional transverse "kick" broadens the p_T^e distribution and washes out the sharp Jacobian edge. This smearing behaviour is further illustrated in Appendix E.

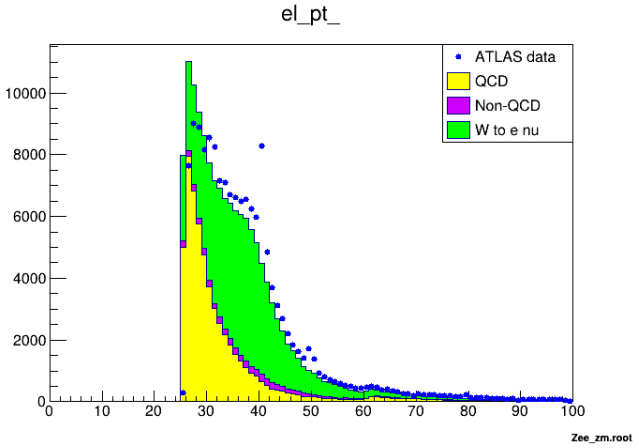


FIG. 3: Transverse momentum distribution of the electron (p_T^e on the x-axis in GeV) for ATLAS data compared to simulated signal and background. This kinematic observable exhibits the characteristic Jacobian peak and serves as the primary distribution for extracting the W-boson mass. The data is shown here prior to applying the final stringent kinematic cut selection, illustrating the substantial QCD and secondary background contributions that must be mitigated to reliably isolate the signal.

b. ***Electron isolation (`el_etiso < 4`):*** The calorimeter isolation variable `el_etiso` measures the transverse energy deposited in a cone of $\Delta R < 0.45$ around the electron direction [2]. Electrons from prompt W decays [3] are largely isolated (spanning roughly 0 GeV to 4 GeV on the x-axis in Figure 5). In contrast, QCD backgrounds often feature electrons from heavy-flavor decays inside jets, or hadronic jets faking an electron, both of which deposit significant surrounding energy, as evidenced by the long tail (`el_etiso` $>$ 4 GeV). Requiring `el_etiso` < 4 GeV effectively filters out these non-isolated candidates while retaining the bulk of the signal.

c. ***Missing transverse energy (`25 < etmis`):*** Requiring $E_T^{\text{miss}} > 25$ GeV retains the peak region of the $W \to e\,\nu$ signal while removing the low-energy regime where both QCD and non-QCD backgrounds dominate (see Figure 6). No upper cut on E_T^{miss} is imposed because the background contributions naturally decrease sufficiently fast at high values.

[2] We note that determining the background scaling visually is inherently subjective. It precludes a rigorous statistical minimization (such as a maximum likelihood fit) and introduces an arbitrary normalization offset. This translates directly into a systematic uncertainty on the final W mass measurement, which is accounted for in Section VI F.

[3] A prompt decay refers to an electron originating directly from the primary hard scattering process (the W boson decay vertex), rather than from delayed, secondary decays of heavy-flavor hadrons.

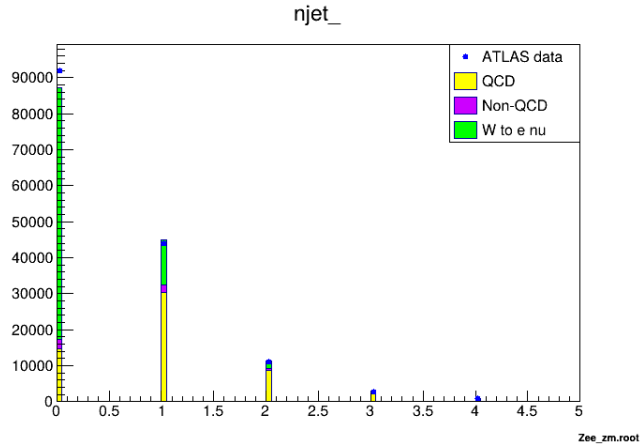


FIG. 4: Distribution of the number of reconstructed jets (`njet`) in ATLAS data overlaid with stacked simulations (the x-axis is in GeV and the y-axis represents counts). The signal-to-background ratio is largest in the exclusive zero-jet bin. Events with one or more jets are rejected because they typically involve the W boson recoiling against initial-state gluon radiation [2], which kinematically smears the electron p_T spectrum and degrades the mass measurement.

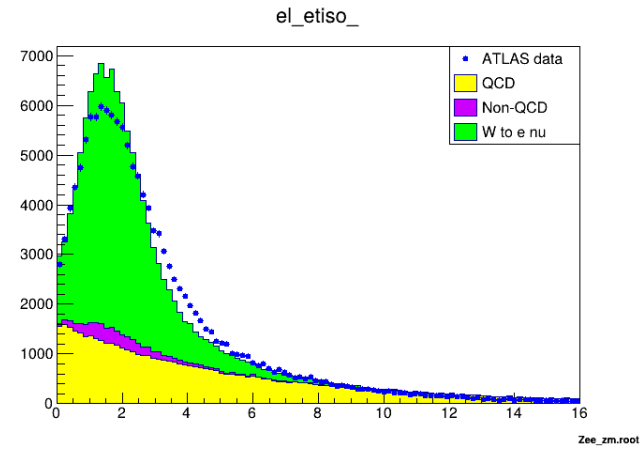


FIG. 5: Distribution of the electron calorimeter isolation variable (`el_etiso`) in ATLAS data and simulation (the x-axis is in GeV and the y-axis represents counts). This variable quantifies the amount of surrounding transverse energy near the electron candidate. The applied threshold at 4 GeV effectively separates the isolated prompt electrons characteristic of W signal decays from the non-isolated electrons typical of heavy-flavor QCD background.

d. W-boson transverse momentum (`ptw < 36`): We impose an upper cut on the reconstructed transverse momentum of the W-boson candidate at $p_T^W < 36\,\text{GeV}$ (see Figure 7). Originally, we intended to use the $Z^0 \to e^+e^-$ Jacobi peak as an explicit cross-check for our calibration.

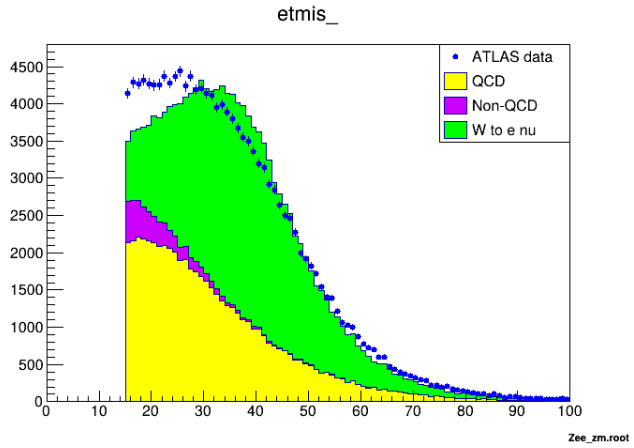


FIG. 6: Distribution of the missing transverse energy (E_T^{miss}, the x-axis is in GeV and the y-axis represents counts) . Missing transverse energy serves as a proxy for the unobserved neutrino emitted in the $W \to e\nu$ decay.

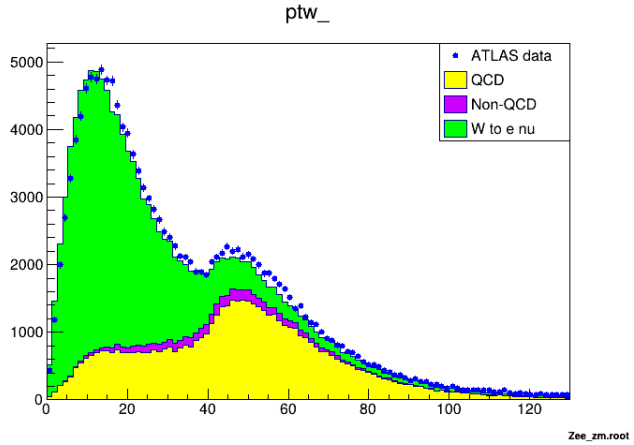


FIG. 7: Distribution of the estimated W-boson transverse momentum (p_T^W, the x-axis is in GeV and the y-axis represents counts). The applied upper cut at 36 GeV suppresses a secondary kinematic peak at higher momentum, drastically increasing the signal-to-background ratio.

However, when we included the Z^0 decay data (by not imposing any cuts on `ptw`), the software failed to achieve simultaneous convergence for both peaks. If the W fits converged, the Z^0 fit exhibited an unphysical discontinuity; conversely, if the Z^0 fit converged, some W fits diverged. So we decided to optimise solely for the W mass by cutting on `ptw` to increase the SBR, which drops significantly for `ptw` $> 36\,\text{GeV}$.

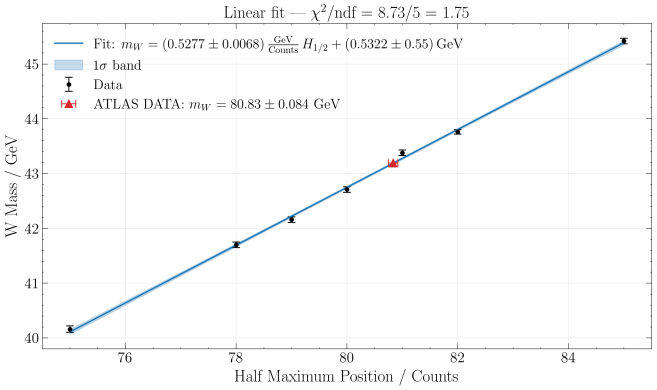


FIG. 8: Gauge curve relating the measured half-maximum point $H_{1/2}$ to the input W-boson mass m_W. The black data points correspond to seven simulated samples. The blue line represents the linear fit used to calibrate the measurement. Errors in $H_{1/2}$ are drawn on the points, though they are barely visible as their relative statistical size is in the per mille range. The red marker designates the extracted mass when applying this curve to the observed ATLAS data.

D. Gauge Curve of the W-Boson Mass

To calibrate the mass measurement, the half-maximum procedure is systematically applied to seven simulated Monte Carlo data sets with known input W-boson masses (see Table VII in Appendix G), as well as to the real recorded ATLAS data. The extracted $H_{1/2}$ values from the simulations serve as calibration points to construct a gauge curve (see Figure 8), which translates the measured $H_{1/2}$ in ATLAS data into a definitive W-boson mass.

For each simulated sample, we perform the half-maximum fit and record the resulting $H_{1/2}$ value. We then fit a linear function

$$H_{1/2} = a \cdot m_W + b \tag{14}$$

$$m_W = \frac{H_{1/2} - b}{a} \tag{15}$$

to the seven $(H_{1/2}\,,\, m_W)$ data points. Equation (15) explicitly serves as the mathematical transfer function used to convert a measured $H_{1/2}$ value into an equivalent W mass.

Applying the raw calibration with $\alpha_{\text{QCD}} = 0.38$ and nominal cuts, the fit yields:

- **Slope (a):** $0.5277(68)\,\text{counts/GeV}$
- **Intercept (b):** $0.5322(5490)\,\text{counts}$
- **Covariance (a, b):** $-0.0037\,\text{counts}^2/\text{GeV}$
- **Goodness of Fit:** $\chi^2/\text{ndf} = 8.72/5 \approx 1.74$

As expected for a linear regression, the fit returns a non-zero covariance between the slope and the y-axis intercept. The goodness-of-fit statistic gives $\chi^2 = 8.72$ for 5 degrees of freedom (ndf).

The value $\chi^2/\text{ndf} \approx 1.74$ likely indicates that the uncertainties on the y-values are somewhat small. We nevertheless use these errors as they are determined from the Jacobi-peak fit and they provide the only available pointwise uncertainty estimate for the calibration curve. In addition, Figure 8 shows a clear linear trend indicating that a linear model properly describes the data.

Nonetheless, using more simulated data sets with different W masses is a natural way to improve the gauge curve.

E. W-Boson Mass in ATLAS data

Using the calibrated gauge curve and Equation (15), the extracted half-maximum for the ATLAS data, $H_{1/2}^{\text{data}} = 43.19(4)\,\text{GeV}$, translates directly into a calculated nominal W-boson mass of $m_W = 80.83(8)\,\text{GeV}$.

Note that the 0.08 GeV error quoted here represents purely the statistical uncertainty propagated from the fits. The systemic biases and systematic uncertainties associated with this central value are quantified and discussed fully in Section VI F.

F. Error Sources

The total uncertainty on the measured W-boson mass comprises both systematic and statistical contributions.

1. *Systematic Uncertainties*

We identify four main sources of systematic uncertainty:

1. **QCD scale factor α_{QCD}:** The normalisation of the QCD background directly affects the shape of the electron p_T distribution and hence the extracted half-maximum point.
2. **Cut selection:** Alternative (looser or tighter) cut selections modify the composition of signal and background in the fitted sample.
3. **Fit range for $H_{1/2}$:** The choice of the lower and upper boundaries of the fit region influences the half-maximum determination.
4. **ECAL energy calibration:** Residual miscalibrations of the electromagnetic calorimeter propagate into the electron p_T measurement. As discussed in Section V A, this effect appears to be negligible at our level of precision.

TABLE I: Summary of systematic parameter variations applied to the extraction of the W-boson mass. Each row defines a unique testing configuration. Parameters kept at their fully calibrated baseline values are indicated as "Nominal". By executing the full gauge-curve fit procedure for each row, we establish bounds on the systematic uncertainty inherent to our analysis choices.

Systematic Label	α_{QCD}	Cut Selection	Fitting Range
Fully Calibrated	0.38	Nominal*	30–60
QCD Factor: 0.30	0.30	Nominal	30–60
QCD Factor: 0.55	0.55	Nominal	30–60
Cut Selection: Loose	0.38	Loose**	30–60
Cut Selection: Tight	0.38	Tight***	30–60
Fit Range: 25–70	0.38	Nominal	25–70
Fit Range: 32–55	0.38	Nominal	32–55

* `njet == 0 && el_etiso < 4 && ptw < 36 && 25 < etmis`

** `njet == 0 && el_etiso < 6 && ptw < 46 && 20 < etmis`

*** `njet == 0 && 0.5 < el_etiso < 3.5 && ptw < 25 && 30 < etmis`

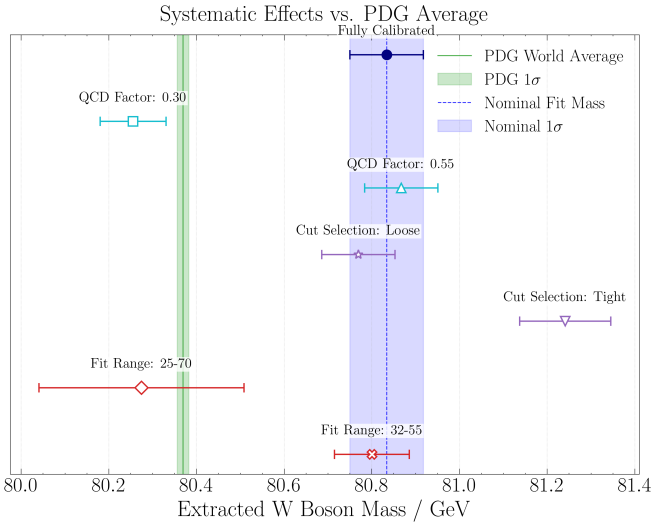


FIG. 9: Visual overview of systematic effects on the extracted W-boson mass measurement. The solid blue line represents the nominal, fully calibrated mass measurement, with its corresponding 1σ statistical error band. Each marker displays the central mass value obtained when running the analysis under a specific alternative parameter variation. Comparing these shifts to the Particle Data Group [1] (PDG) world average (green band) provides a comprehensive view of the systematic sensitivity of the measurement.

Table I summarises the variations applied to quantify each systematic effect in isolation. To quantify the overall systematic uncertainty, we evaluate the shift of the central W-boson mass value for each variation relative to the fully calibrated nominal measurement (visualized in Figure 9).

Because the upward and downward variations of the parameters yield highly asymmetric shifts in the extracted mass, we calculate separate positive ($+\sigma_{\mathrm{sys}}$) and negative ($-\sigma_{\mathrm{sys}}$) uncertainties. The individual mass shifts Δm_W for each variation are categorized by their sign, as detailed in Table II.

TABLE II: Signed shifts in the extracted W-boson mass relative to the nominal configuration ($m_W = 80.83\,\mathrm{GeV}$). Positive shifts are combined in quadrature to form the upward systematic uncertainty, and negative shifts are combined to form the downward systematic uncertainty. If both variations of a single category yield a shift in the same direction (e.g., the Fit Range), only the maximum shift is used.

Systematic Source	$+\Delta m_W$ [GeV]	$-\Delta m_W$ [GeV]
QCD Scale Factor		
$\alpha_{\mathrm{QCD}} = 0.30$	–	−0.58
$\alpha_{\mathrm{QCD}} = 0.55$	0.03	–
Cut Selection		
Loose Cuts	–	−0.06
Tight Cuts	0.41	–
Fit Range		
25–70	–	−0.56
32–55	–	−0.03
Quadrature Sum σ_{sys}	+0.41	−0.81

As visualised in Figure 9 and quantified in Table II, the magnitudes of the systematic variations differ significantly.

The individual sources of systematic uncertainty, ranked by their maximum impact on the measurement, are listed below:

1. **QCD Background Scaling (Maximum deviation: $-0.58\,\mathrm{GeV}$):** The most severe systematic shift occurs when the QCD scale factor is lowered to $\alpha_{\mathrm{QCD}} = 0.30$. Conversely, overestimating the background contribution ($\alpha_{\mathrm{QCD}} = 0.55$) has a comparatively negligible effect, yielding a minor shift of only $0.03\,\mathrm{GeV}$.

2. **Fit Range Selection (Maximum deviation: $-0.56\,\mathrm{GeV}$):** The second largest deviation is induced by extending the kinematic fit window from 30–60 GeV to 25–70 GeV. This variation forces the mathematical algorithm to encompass a larger region of the spectrum, requiring it to simultaneously fit and explain more data points along both the low-p_T and high-p_T tails. This boundary expansion shifts the slope of the fit and subsequently the extracted half-maximum point downwards. Conversely, narrowing the fit range to 32–55 GeV avoids

this broadening effect and yields only a small shift ($-0.03\,\mathrm{GeV}$), indicating that the instability is primarily driven by the extended boundaries.

3. **Kinematic Cut Selection (Maximum deviation: $0.41\,\mathrm{GeV}$):** Applying the highly stringent "Tight" cut selection induces an upward shift. On the other hand, relaxing the selection criteria ("Loose" cut) introduces only a minor downward variation ($-0.06\,\mathrm{GeV}$). This asymmetry suggests that while restricting the phase space too heavily introduces strong biases, slightly loosening the nominal cuts does not severely alter the kinematic composition of the fitted sample.

To extract a final uncertainty from these independent effects, they must be statistically combined. When combining these uncertainties, we neglect the individual possible correlations between the different systematic sources. This has been a deliberate choice, as calculating the true covariance matrix across all phase-space variations requires a statistical analysis well beyond the time available for this laboratory course. For each systematic source, positive mass shifts are added in quadrature to form the total upward uncertainty, and negative shifts are added in quadrature to form the total downward uncertainty. Under the standard assumption of statistically independent errors, the asymmetric systematic uncertainties are calculated as:

$$+\sigma_{\mathrm{sys}} = \sqrt{(0.03)^2 + (0.41)^2} \approx 0.41\,\mathrm{GeV}\,, \tag{16}$$

$$-\sigma_{\mathrm{sys}} = -\sqrt{(-0.58)^2 + (-0.06)^2 + (-0.56)^2} \approx -0.81\,\mathrm{GeV}\,. \tag{17}$$

Furthermore, we deliberately do not use the statistical errors of the individual mass shifts (which stem from the inherent fit uncertainties of the Jacobi-peak and gauge-curve algorithms) when calculating the systematic error. Doing so would incorrectly mix statistical fluctuations with systematic biases.

For instance, we note that in the $\alpha_{\mathrm{QCD}} = 0.30$ variation, the $m_W = 75\,\mathrm{GeV}$ simulated dataset exhibited an algorithmic fit uncertainty of $H_{1/2} = 39.99(49680994)\,\mathrm{GeV}$. This error, though mathematically enormous, merely inflates the χ^2 and reduces the statistical weight of this specific point in the linear gauge fit; it has limited impact on the final extracted W mass. Specifically, manually restricting this error to a typical error size [4] of $dH_{1/2} = 0.05\,\mathrm{GeV}$ decreases the observed W mass shift from $0.5785\,\mathrm{GeV}$ to $0.5673\,\mathrm{GeV}$, corresponding to roughly a $2\,\%$ deviation. Therefore, the impact of this error on the systematic mass shift is limited.

[4] Typical in the sense that it is equal in magnitude to the other errors of the Jacobi-peak fits (see Figure 15 in Appendix F).

2. *Statistical Uncertainty and Final Result*

The dominant statistical uncertainty originates from the initial Jacobi-peak fit applied to the data, as well as the linear fit of the gauge curve. Because this uncertainty is driven by the finite size of the datasets and is expected to decrease with additional simulated Monte Carlo events (i.e., more m_W hypotheses or higher event counts), we classify it strictly as statistical rather than systematic.

Strictly speaking, the calculated systematic shifts themselves discussed in Section VI F 1 also contain statistical uncertainties due to limited simulation sizes, as suggested by the error bars in Figure 9. Disentangling these from the true systematic biases, though, is beyond the scope of this analysis.

The statistical uncertainty is calculated via standard Gaussian error propagation of Equation (15), incorporating the uncertainty in the measured half-maximum $H_{1/2}^{\mathrm{data}}$, the errors in the slope a, the y-axis intercept b, and their correlation $\mathrm{cov}(a, b)$. The resulting propagated statistical error is $0.08\,\mathrm{GeV}$.

We find that this statistical uncertainty is roughly 5 to 10 times smaller than the calculated asymmetric systematic bounds ($^{+0.41}_{-0.81}\,\mathrm{GeV}$). This stark contrast highlights that this experiment is entirely systematic-limited.

The final measured mass of the W boson from ATLAS data, including both statistical and asymmetric systematic uncertainties, is determined to be:

$$m_W = 80.83 \pm 0.08\ (\mathrm{stat.})\ ^{+0.41}_{-0.81}\ (\mathrm{sys.})\,\mathrm{GeV}\,. \tag{18}$$

To evaluate the compatibility of our measurement with the accepted Standard Model value, we calculate the z-score relative to the PDG world average of $m_W^{\mathrm{PDG}} = 80.3692(133)\,\mathrm{GeV}$ [1].

Because the nominal measured mass ($80.83\,\mathrm{GeV}$) is higher than the world average, we must utilize the downward (negative) systematic uncertainty bound to evaluate the statistical tension. We first combine our statistical and negative systematic uncertainties in quadrature to obtain a total downward measurement uncertainty of $\sigma_{\mathrm{tot},-} = \sqrt{0.08^2 + 0.81^2} \approx 0.814\,\mathrm{GeV}$. The z-score is then given by:

$$z = \frac{|80.83 - 80.3692|}{\sqrt{0.08^2 + 0.81^2 + 0.0133^2}} = \frac{0.4608}{0.814} \approx 0.57\,. \tag{19}$$

A deviation of roughly 0.57σ indicates that our measured W-boson mass is in excellent statistical agreement with the current world average, despite the asymmetric nature of our systematic bounds.

VII. CONCLUSION

In conclusion, after calibrating the ECAL using the Z-Boson Peak, our measurement demonstrates that the Jacobian peak method effectively extracts the W-boson mass. By optimizing the signal-to-background ratio

through stringent kinematic cuts on jet multiplicity, missing transverse energy, electron isolation and transverse momentum of the W boson, we isolated the $W \to e\nu$ signal in ATLAS data.

While technical limitations within the provided laboratory software, namely the not achieving simultaneous convergence of both the Z^0 Jacobi-peak fit as well as all the W Jacobi-peak fits, hindering a cross-check of the analysis, the conservative, robust estimation of systematic uncertainties ensures the physical validity of the final result. In Sections VI D and VI F , we derived a final W-boson mass of

$$m_W = 80.83 \pm 0.08 \text{ (stat.)} \pm^{+0.41}_{-0.81} \text{ (sys.) GeV}$$

which lies within 0.51σ of the PDG world average of $m_W^{\text{PDG}} = 80.3692(133)\,\text{GeV}$ [1] . The fact that the central value is shifted by approximately 0.46 GeV but remains well within the large systematic bounds underscores the importance of accurately calibrating the QCD background scaling factor as well as choosing sensible cuts. In addition, using more simulated data sets for the W mass would lead to a more accurate gauge curve and thereby a more accurate W mass.

In a separate assignment, the analysis of mystery event displays with dilepton final states identified heavy-quark pair production, either $t\bar{t}$ or direct $b\bar{b}$, as the most consistent hypothesis, based on the observation of opposite-charge, mixed-flavour dileptons, hadronic jets with closely-spaced tracks, and variable E_T^{miss}.

ACKNOWLEDGMENTS

We acknowledge the usage of Gemini 3.1 [4] for drafting support, text refinement, and LaTeX formatting assistance during the preparation of this manuscript.

Appendix A: ECAL Cluster Energy vs. Inner-Detector Track Momentum

TABLE III: ECAL cluster energy and Inner-Detector track momentum for 20 electron events. Events marked with † are identified as outliers (very low $|p_T|$, unreliable curvature measurement) and excluded from the summary statistics.

Event	$\lvert p_T\rvert$ [GeV]	E_T [GeV]	E/p
1	25.38 ± 1.111	31.7	1.249 ± 0.055
2†	6.03 ± 0.970	45.2	7.496 ± 1.206
3	41.90 ± 1.494	52.4	1.251 ± 0.045
4†	3.14 ± 0.410	45.8	14.586 ± 1.905
5	44.33 ± 1.859	37.1	0.837 ± 0.035
6	35.00 ± 1.063	29.5	0.843 ± 0.026
7	36.75 ± 0.594	30.1	0.819 ± 0.013
8	50.87 ± 1.335	49.2	0.967 ± 0.025
9	47.35 ± 3.302	49.0	1.035 ± 0.072
10	35.07 ± 1.153	35.1	1.001 ± 0.033
11	26.52 ± 0.680	26.0	0.980 ± 0.025
12	52.03 ± 1.009	45.2	0.869 ± 0.017
13	25.07 ± 0.530	49.0	1.955 ± 0.041
14†	6.64 ± 0.331	29.7	4.473 ± 0.223
15	49.15 ± 2.243	43.2	0.879 ± 0.040
16	30.54 ± 1.602	55.4	1.814 ± 0.095
17	36.24 ± 0.637	33.6	0.927 ± 0.016
18	55.42 ± 1.159	48.7	0.879 ± 0.018
19	20.80 ± 0.467	32.4	1.558 ± 0.035
20	37.99 ± 2.212	53.2	1.400 ± 0.082
Mean E/p (all 20 events)			2.291 ± 0.721
Std. dev. (all 20 events)			3.224
Mean E/p (excl. outliers, $N = 17$)			1.133 ± 0.083
Std. dev. (excl. outliers)			0.344

Appendix B: Event-by-Event Observations for Assignment 7

In all events, the two leptons carried opposite charge and there were jets present unless explicitly stated otherwise.

- **Event 1:** ($\sum E_T = 391\,\text{GeV}$, $E_T^{\text{miss}} = 53\,\text{GeV}$) showed back-to-back leptons accompanied by back-to-back hadronic jets, with a total transverse momentum of roughly 220–230 GeV.
- **Event 2:** ($\sum E_T = 353\,\text{GeV}$, $E_T^{\text{miss}} = 2.6\,\text{GeV}$) contained only two lepton tracks: a muon with $|p_T| = 33\,\text{GeV}$ and an electron with $|p_T| = 52\,\text{GeV}$. The missing transverse energy was anomalously low given the total energy in the event without any jets.
- **Event 3:** ($\sum E_T = 520\,\text{GeV}$, $E_T^{\text{miss}} = 164\,\text{GeV}$) featured one muon ($|p_T| = 87\,\text{GeV}$) and a hadronic jet cluster carrying roughly 240 GeV with four tracks in total. The combination of two different lepton flavours and energetic jets is suggestive of heavy-quark production.
- **Event 5:** ($\sum E_T = 190\,\text{GeV}$, $E_T^{\text{miss}} = 12\,\text{GeV}$) contained two energetic muons ($|p_T| = 12$ and 64 GeV) plus jets. The comparatively low $\sum E_T$ makes $t\bar{t}$ production unlikely for this particular event.
- **Event 10a:** ($\sum E_T = 225\,\text{GeV}$, $E_T^{\text{miss}} = 50\,\text{GeV}$) showed two muons and one jet.
- **Event 10b:** ($\sum E_T = 390\,\text{GeV}$, $E_T^{\text{miss}} = 68.7\,\text{GeV}$) also contained two muons, with $|p_T| = 21$ and 39 GeV.
- **Event 12:** ($\sum E_T = 468\,\text{GeV}$, $E_T^{\text{miss}} = 154\,\text{GeV}$) contained one muon and one electron.
- **Event 45a:** ($\sum E_T = 696\,\text{GeV}$, $E_T^{\text{miss}} = 80\,\text{GeV}$) was the most energetic event in the sample. It contained one muon ($|p_T| = 25\,\text{GeV}$), one electron ($|p_T| = 35\,\text{GeV}$), a hadronic jet with $|p_T| = 142\,\text{GeV}$, and one additional track with $|p_T| = 218\,\text{GeV}$.

- **Event 45b:** ($\sum E_T = 234\,\text{GeV}$, $E_T^{\text{miss}} = 54\,\text{GeV}$) contained one muon and one electron.

Appendix C: Cross sections

Cross sections of the candidate processes for the mystery particle discussed in Section IV .

TABLE IV: Cross-sections of important processes at the LHC in femtobarn, spanning roughly 13 orders of magnitude. From [2].

Process	σ [fb]
Total cross-section	ca. 10^{14}
b-quark pairs	0.5×10^{12}
Dijets with a $p_T > 100\,\text{GeV}$	ca. 1.8×10^{9}
W-boson production	0.17×10^{9}
W-boson production (decay of W into e ν)	17×10^{6}
Z^0-boson production (decay of Z into e e)	1.6×10^{6}
top-quark pair production	8×10^{5}
Z^0 pair production	1.6×10^{3}
Z^0 pair production ($Z^0 \to e^+e^-$ or $Z^0 \to \mu^+\mu^-$)	70
Higgs production ($m_H = 130\,\text{GeV}$)	35×10^{3}
Higgs production ($m_H = 300\,\text{GeV}$)	12×10^{3}
Higgs prod. ($m_H = 300\,\text{GeV}$), ($H \to Z^0Z^0$, where $Z^0 \to e^+e^-$ or $Z^0 \to \mu^+\mu^-$)	16

Appendix D: Energy Calibration correction factors

Energy calibration factors for the ECAL detector used in the Z peak calibration, done in Section V .

TABLE V: Energy correction factors as a function of p_T.

p_T range [GeV]	Correction factor
0–10	0.9674
10–20	1.0324
20–30	1.0378
30–40	1.0276
40–50	1.0078
50–60	0.9884
60–70	0.9894
70–80	0.9953
80–90	0.9906
90–100	0.9942

Appendix E: p_T^W Jet-Multiplicity Dependence

This section documents the reconstructed W-boson transverse-momentum (p_T^W) distribution after applying the QCD scale factor, separated by exact integer jet multiplicities (zero through four), as shown in Figures 10–14.

TABLE VI: Energy correction factors as a function of η.

η range	Correction factor
−2.49 to −2.4	1.0700
−2.4 to −2.3	1.0678
−2.3 to −2.2	1.0531
−1.5 to −1.0	1.0054
−1.0 to −0.5	1.0008
−0.5 to 0.0	0.9976
0.0 to 0.5	0.9971
0.5 to 1.0	1.0017
1.0 to 1.5	1.0045
2.2 to 2.3	1.0508
2.3 to 2.4	1.0481
2.4 to 2.51	1.0359

As the jet multiplicity increases, the p_T^W distribution broadens and its central peak shifts to higher values. This physically reflects the recoil of the W boson caused by initial-state radiation [2] and additional hard hadronic activity in the detector. This progressive smearing behaviour strongly motivates the stringent `njet == 0` requirement applied in our final cut selection (see Section VI C) to preserve the integrity of the Jacobian fit.

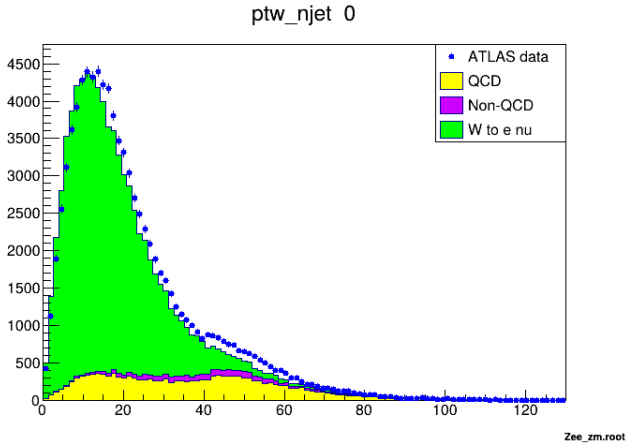


FIG. 10: p_T^W distribution for events with exactly zero reconstructed jets, after applying the QCD scale-factor calibration. The sharply defined peak at low p_T^W confirms that the W boson is produced with negligible transverse boost in exclusive zero-jet events. This pure signal region provides the sharpest Jacobian edge for the electron p_T spectrum fit.

Appendix F: Jacobi Peak Fits

This section provides visual examples of the empirical fits used to extract the half-maximum point ($H_{1/2}$) of the Jacobian peak using a fully calibrated QCD-background scaling factor as well as cut selection. As discussed in Section VI A, rather than using the absolute peak maximum,

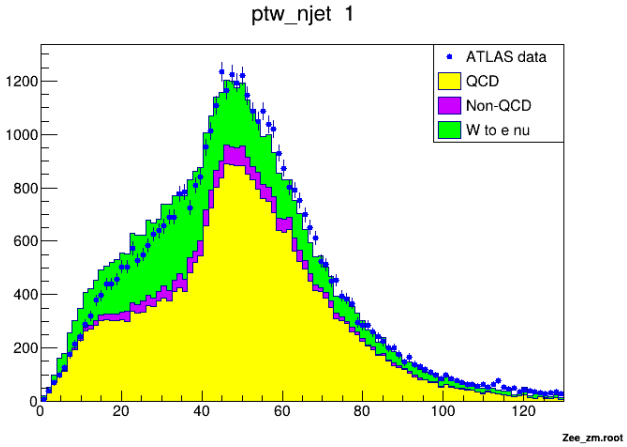


FIG. 11: p_T^W distribution for events with exactly one reconstructed jet. The peak is broader and notably shifted to higher p_T^W values compared to the zero-jet case. This shift physically represents the W boson acquiring transverse momentum as it recoils against initial-state gluon radiation (the associated jet).

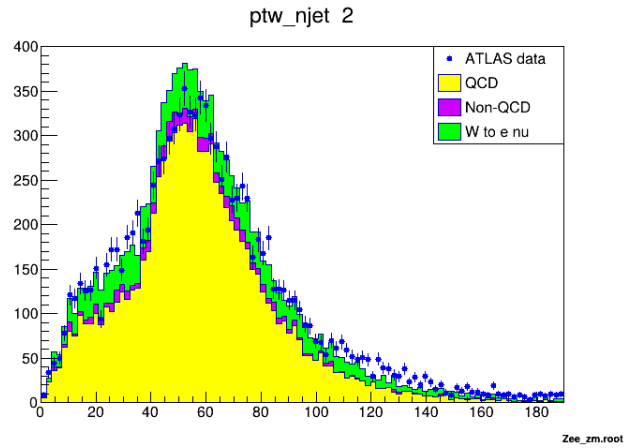


FIG. 12: p_T^W distribution for events with exactly two reconstructed jets. The kinematic smearing effect continues to increase. The higher transverse momentum of the W boson severely broadens the electron p_T spectrum, decreasing the viability of a precise half-maximum fit.

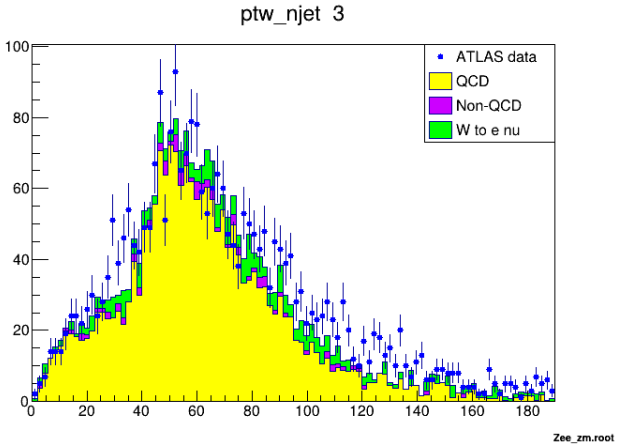


FIG. 13: p_T^W distribution for events with exactly three reconstructed jets. Both the kinematic recoil shift and the overall QCD background levels rise significantly as the event becomes highly hadronic.

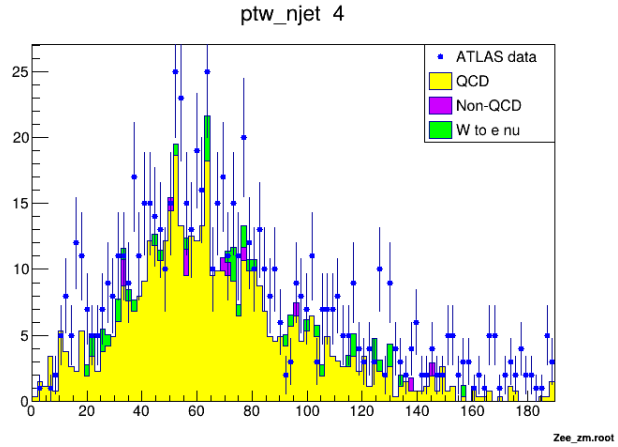


FIG. 14: p_T^W distribution for events with exactly four reconstructed jets. The transverse momentum distribution is heavily broadened and almost uniform compared to the sharp peak of the zero-jet case. Furthermore, the signal-to-background ratio is substantially degraded, rendering this phase space entirely unusable for a reliable Jacobian peak measurement.

the falling edge of the electron transverse-momentum spectrum is fitted to reliably find $H_{1/2}$. This mathematical procedure is executed iteratively for the ATLAS data as well as for each of the simulated Monte Carlo mass hypotheses to construct the linear calibration gauge curve.

Appendix G: Simulated data sets used for the gauge curve

To accurately calibrate the relationship between the measured half-maximum point $H_{1/2}$ and the true W-boson mass, the analysis relies on a specifically generated suite of simulated Monte Carlo (MC) data sets. Table VII lists all data samples utilized in this study, including the actual ATLAS collision data and the corresponding MC signal samples. Each simulated signal sample represents a different assumed W-boson mass.

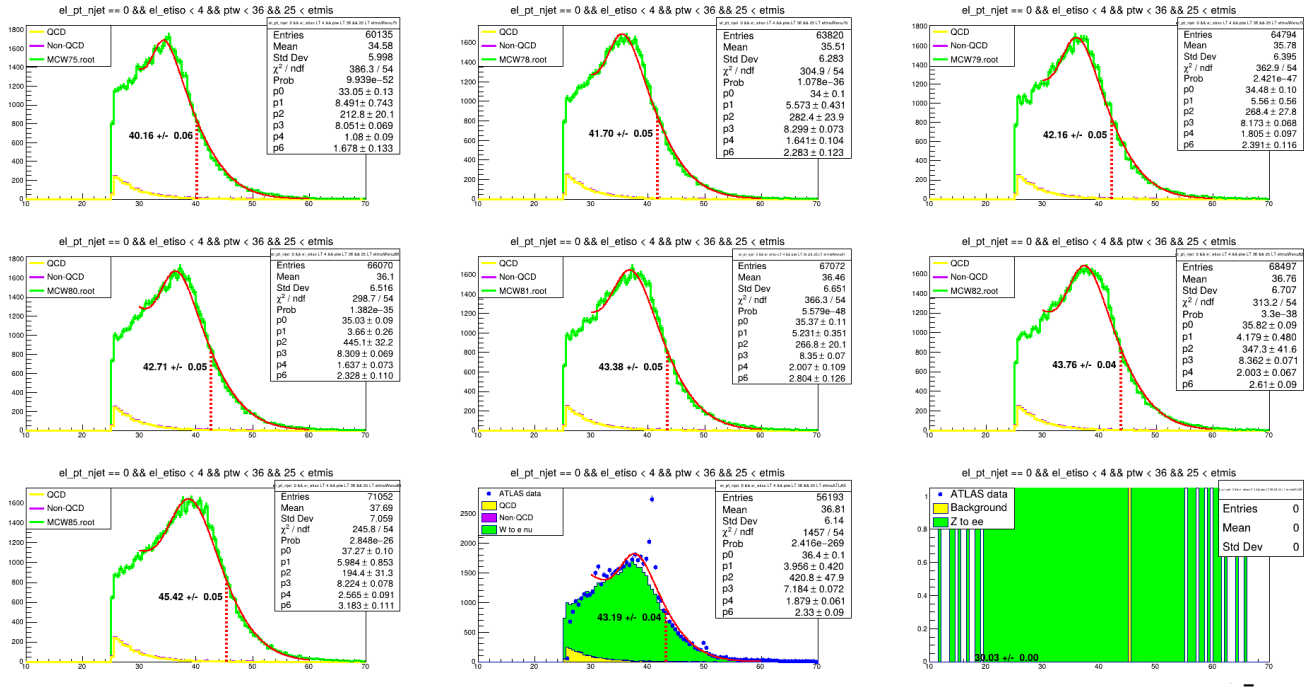


FIG. 15: Representative fits of the Jacobian peak in the electron p_T spectra. The plots show the empirical function fitted to the falling edge of the p_T^e distribution within the nominal kinematic window of 30–60 GeV. These specific fits are used to accurately extract the half-maximum point $H_{1/2}$. The bottom right panel represents the the Z^0 Jacobi-peak which is not visible because of the globally applied `ptw` cut (also applied to the Z^0). The reason for the `ptw` cut is explained in Section VI C

TABLE VII: Summary of the simulated Monte Carlo $W \to e\nu$ data sets used to construct the gauge curve. Each sample is generated with a different assumed W-boson mass, allowing us to map the measured half-maximum point $H_{1/2}$ to a definitive invariant mass.

Dataset	Abbreviation	Description
$W \to e\nu$	`Wenu`	ATLAS data with $W \to e\nu$ candidates
$Z \to ee$	`Zee`	ATLAS data with $Z \to ee$ candidates
$W \to e\nu$ sim.	`MCW75`	simulated signal, $m_W = 75$ GeV
$W \to e\nu$ sim.	`MCW78`	simulated signal, $m_W = 78$ GeV
$W \to e\nu$ sim.	`MCW79`	simulated signal, $m_W = 79$ GeV
$W \to e\nu$ sim.	`MCW80`	simulated signal, $m_W = 80$ GeV
$W \to e\nu$ sim.	`MCW81`	simulated signal, $m_W = 81$ GeV
$W \to e\nu$ sim.	`MCW82`	simulated signal, $m_W = 82$ GeV
$W \to e\nu$ sim.	`MCW85`	simulated signal, $m_W = 85$ GeV

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